THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. 3500 m expressed in km is: 1

(a) 3.5 km **(b)** 350 km **(c)** 0.35 km **(d)** 35 km

2. Which of the following pairs includes only soluble substances? 1

(a) sand and salt **(b)** salt and sugar **(c)** sand and chalk **(d)** chalk and sugar

3. Mercury level before measuring should be: 1

(a) below 35 °C **(b)** at 40 °C **(c)** at 42 °C **(d)** at 100 °C

4. A beaker of water is placed on a heater. After some time bubbles form and water level falls. This is: 1

(a) boiling **(b)** freezing **(c)** condensation **(d)** melting

5. After removing iron and adding water to the mixture of chalk and salt, filtering gives chalk as: 1

(a) solute **(b)** filtrate **(c)** solvent **(d)** residue

6. A pebble kept in water for days does not change but a seed swells and sprouts. This shows that: 1

(a) seed is living **(b)** seed is lighter
(c) water is hot **(d)** pebble is heavier

7. Which gas in the air supports burning? 1

(a) argon **(b)** nitrogen **(c)** oxygen **(d)** carbon dioxide

8. Which is the correct pair? 1

(a) Mercury: largest planet **(b)** Jupiter: smallest
(c) Venus: hottest planet **(d)** Mars: ringed

9. Which of these helps make the measurement of a pencil's length more accurate? 1

(a) placing the ruler's zero mark at one end and reading straight on
(b) guessing
(c) reading from the side
(d) measuring from a broken end

10. Why do animals living in snowy regions have thick fur or fat? 1

(a) to fly **(b)** to swim better
(c) to reduce heat loss **(d)** to find food

11. Which of the following is a protective food? 1

(a) spinach **(b)** white rice **(c)** oil **(d)** sugar

12. A student says 'All metals are attracted by a magnet'. Which metal shows this is wrong? 1

(a) iron **(b)** cobalt **(c)** aluminium **(d)** nickel

13. A body moving along a circular track at constant speed shows: 1

(a) rest **(b)** circular motion **(c)** no motion **(d)** linear motion

14. Which test would show that a metal rod is a good conductor of heat? 1

(a) wax melts quickly along the rod when one end is heated
(b) it dissolves
(c) it floats
(d) it breaks

15. Why should a lab thermometer not be held by the bulb while reading? 1

(a) it will break **(b)** bulb is sensitive to body heat
(c) it is slippery **(d)** it is cold

16. Which of these is correctly matched? 1

(a) steam – liquid **(b)** ice – solid **(c)** water – gas **(d)** vapour – solid

17. When a hot saturated solution cools, the excess solute: 1

(a) evaporates **(b)** crystallises out **(c)** becomes gas **(d)** dissolves more

18. Frogs lay their eggs in: 1

(a) snow **(b)** sand **(c)** water **(d)** trees

19. Which of the following is recycled? 1

(a) aluminium cans **(b)** smoke **(c)** air **(d)** ash

20. A day on a planet is the time it takes to: 1

(a) rotate once on its axis **(b)** cross
(c) revolve once **(d)** orbit

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. How does a telescope help astronomers? 2

22. Why do fish have scales on their bodies? 2

23. How does a lack of vitamin A affect vision? 2

24. What would happen if two south poles are brought together? 2

25. Distinguish between rotation and revolution with an example. 2

26. Why is a transparent glass not used for bathroom windows? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A pendulum takes 42 s for 21 oscillations. Find its time period and the number of oscillations in 1 minute. 3

UNSTEP MIND

28. Give reasons: 3

(a) a doormat is rough;

(b) a tabletop is smooth;

(c) a window is transparent.

29. Why are the walls of a thermos flask silvered? 3

30. Why do we get fresh water from rain though sea water is salty? 3

31. Explain why sea water is evaporated in shallow pans instead of deep ones. 3

OR

Explain why plants do not need food from outside while animals do. 3

32. Explain how plastics harm the environment. 3

OR

Explain why the Moon is called a natural satellite. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Explain what happens when hot water and cold water are mixed. 5

34. Describe how to prepare drinking water in an emergency using boiling and filtering. 5

35. Analyse how the separation methods used in cooking and farming are related. 5

36. Differentiate between respiration in plants, fish and humans. 5

OR

Explain how the demand for energy has increased and what we can do. 5

37. Explain why the Earth is the only known planet with life. 5

OR

Explain how a hypothesis is formed, tested and accepted or rejected. Use the example of 'Plants need sunlight'. 5

38. Explain how leaves and roots differ between a mango tree and a grass plant. 5

OR

Describe an experiment to test for starch and fat in potato and butter and record the result. 5

END OF QUESTION PAPER — Total: 80 Marks

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